

23,262 DEVELOPERS SURVEYED

# State of Developer Ecosystem 2024:

## Language Trends, Satisfaction, and Strategic Shifts

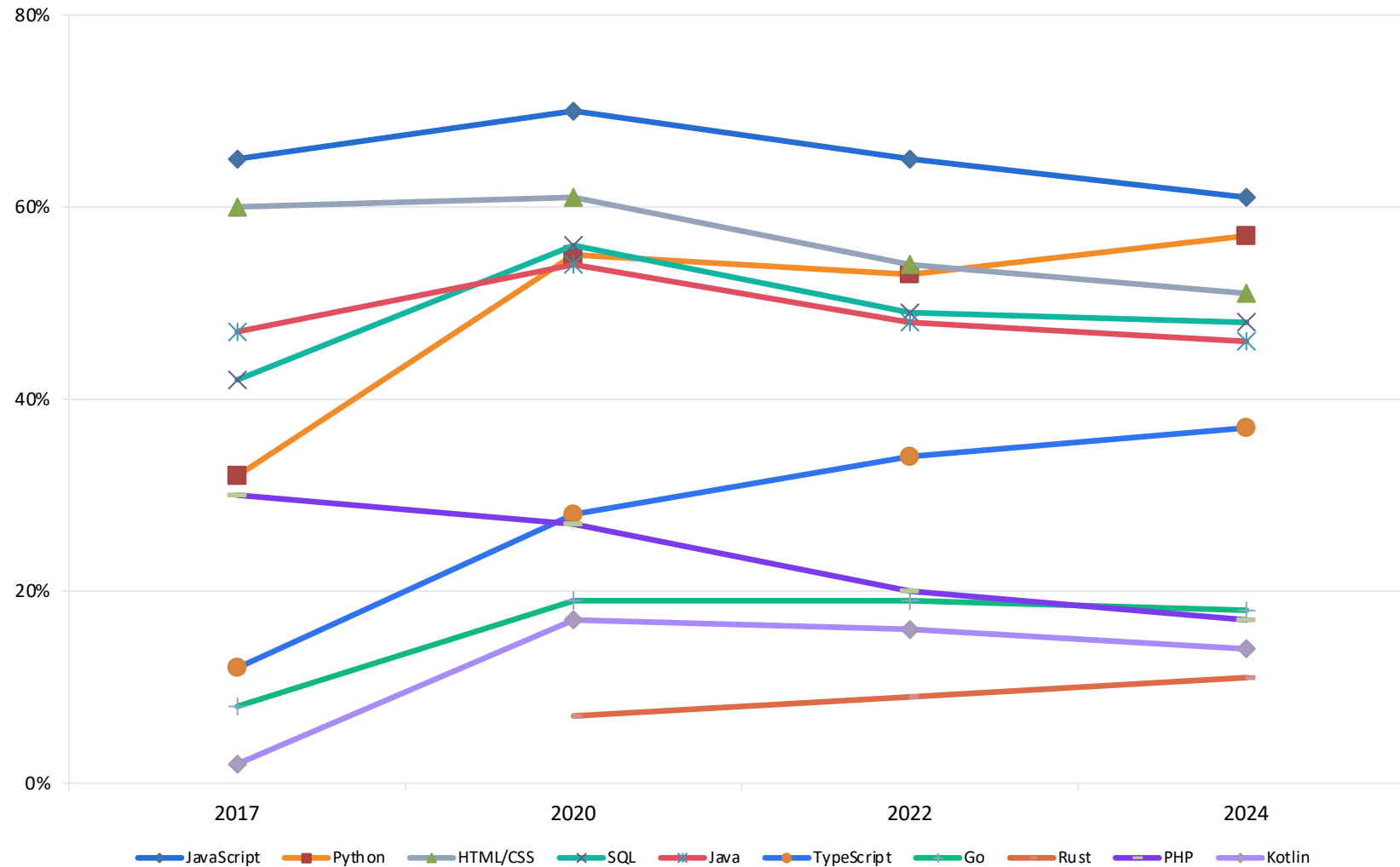
Key insights from JetBrains' annual survey on programming language usage, satisfaction, adoption intent, and domain specialization trends from 2017 to 2024.

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Prepared for engineering leaders, CTOs, and developer experience teams

# Python Closes the Gap: 8 Years of Language Usage Shifts

Percentage of developers using each language | 2017 – 2024

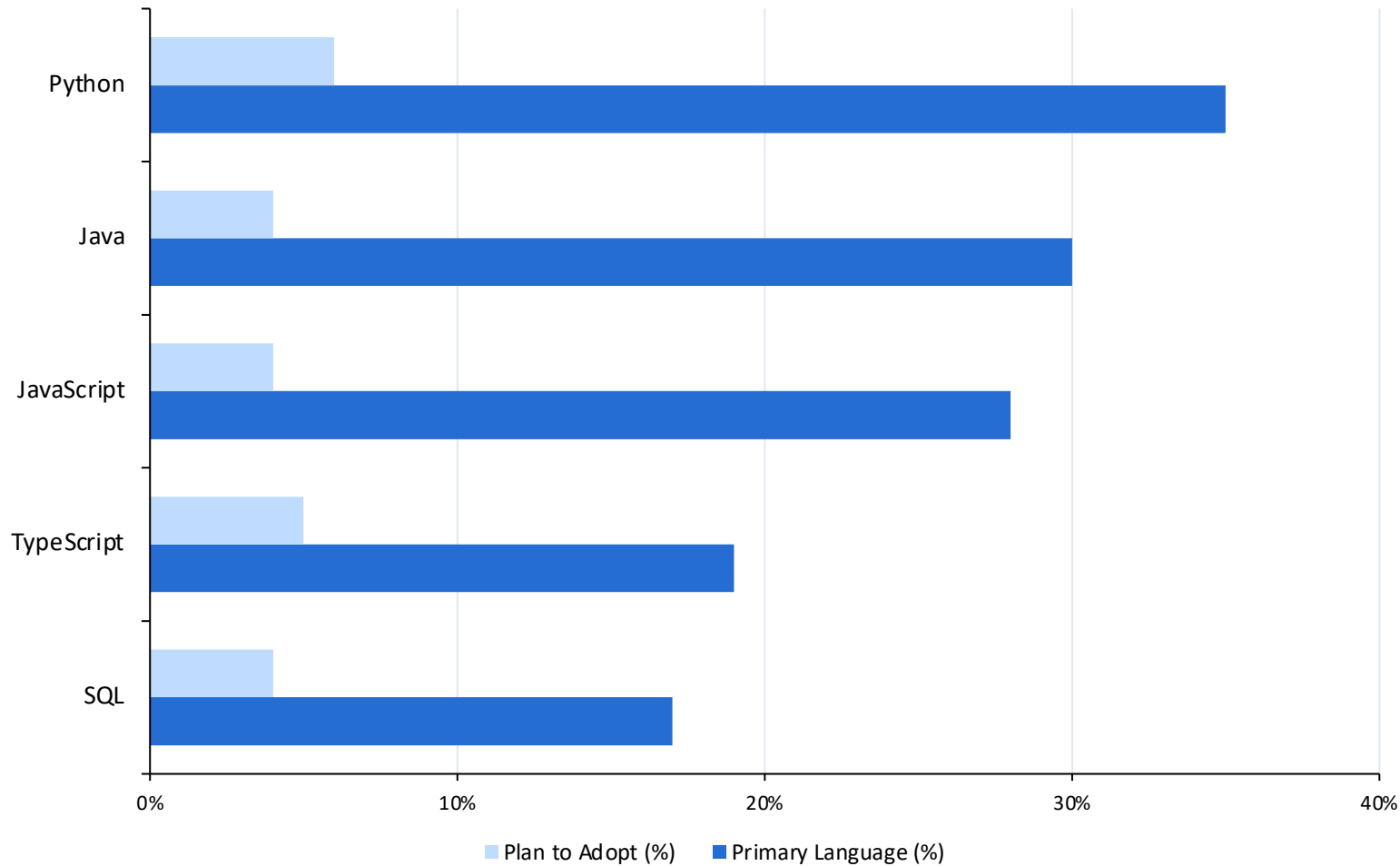


## KEY SHIFTS

- ▲ Python: 32% → 57%  
Nearly doubled in 8 years; AI/ML tailwind
- ▼ JavaScript: 70% → 61%  
Down from 2020 peak but still #1 overall
- ▲ TypeScript: 12% → 37%  
Tripled share; type-safety demand rising
- ▼ PHP: 30% → 17%  
Significant long-term decline continues
- ▲ Rust: ~0% → 11%  
Steady growth in systems programming

# Python Is Now the #1 Primary Language at 35%

Primary language selection vs. planned adoption | Developers can select one primary language



## THE LOCK-IN EFFECT

Adoption intent remains below 6% for every language — developers rarely switch stacks.

This means:

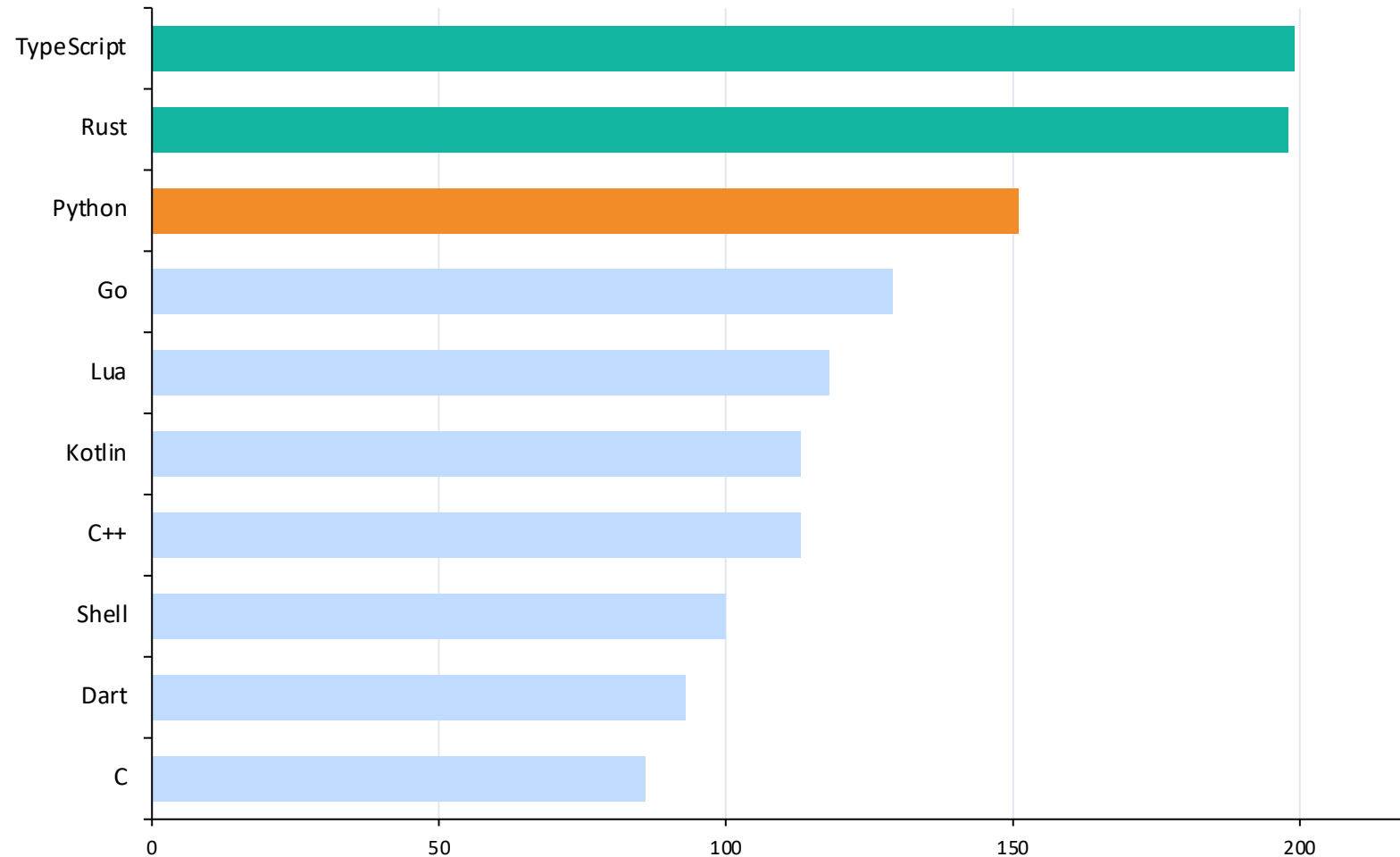
- Hiring must target existing expertise pools
- Ecosystem investment compounds over time
- Migration costs are the real barrier, not language features

Python's #1 primary position reflects the AI/ML boom pulling developers into its ecosystem.

# TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198

Language satisfaction score | Higher = more satisfied developers

## Satisfaction Score



### WHAT DRIVES SATISFACTION

The top-rated languages share common traits:

- Strong type systems (TypeScript, Rust)
- Modern tooling & developer experience
- Expressive, safety-oriented design

Implication for retention strategy:

Teams using high-satisfaction languages report better developer experience, which correlates with lower attrition.

# Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Dominant language per application domain | Share = % of developers selecting that language for the domain

| Domain                        | Lead Language | Share | Strength Indicator     |
|-------------------------------|---------------|-------|------------------------|
| AI / Machine Learning         | Python        | 31%   | <div><div></div></div> |
| Data Science & Analytics      | Python        | 28%   | <div><div></div></div> |
| Data Processing & Analytics   | Python        | 17%   | <div><div></div></div> |
| User Interface                | TypeScript    | 62%   | <div><div></div></div> |
| User Interface                | JavaScript    | 58%   | <div><div></div></div> |
| APIs & Services               | Java          | 49%   | <div><div></div></div> |
| Application Logic & Workflows | Java          | 55%   | <div><div></div></div> |
| Integrating with APIs         | Python        | 38%   | <div><div></div></div> |
| System Tools & Components     | C++           | 27%   | <div><div></div></div> |

## DOMAIN-DRIVEN CHOICE

Language choice is domain-driven, not a general preference exercise.

- Python dominates AI/ML & data science
- TS/JS own the entire UI layer
- Java remains king of backend services
- C++ still leads systems programming

Hiring & training plans should align language investment to domain needs.

# What This Means for Your Technology Strategy in 2025

Five actionable recommendations for engineering leaders

01

## Invest in Python for AI/ML

Python is the undisputed leader in AI/ML (31%) and data science (28%). If you're building AI capabilities, Python proficiency is non-negotiable for your team.

02

## TypeScript for Frontend & Full-Stack

TypeScript tripled to 37% usage and leads developer satisfaction (199). It's the future of frontend and increasingly full-stack development.

03

## Rust & Go for Infrastructure

Both languages are growth plays for cloud-native, microservices, and systems programming. Rust (11%) and Go (18%) address modern infrastructure needs where performance matters.

04

## Plan for Shrinking PHP/Ruby Pools

PHP dropped from 30% to 17%; Ruby from 10% to 4%. Hiring pools are shrinking — plan migration timelines or accept higher recruiting costs.

05

## Satisfaction = Retention

Developer satisfaction correlates with type safety and modern tooling. Factor language choice into your retention strategy — happy developers stay longer.